Revisiting Class-Incremental Learning in the Era of Foundation Models: A Gaussian Density Estimation Perspective

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Abstract. Class-Incremental Learning (CIL) is commonly addressed by preventing catastrophic forgetting through regularization, replay buffers, or architectural adaptations. In this work, we revisit this formulation in the context of modern visual foundation models. We argue that once representations are frozen and sufficiently discriminative, the main challenge of CIL shifts from protecting model parameters to modeling class distributions in a stationary feature space. Building on this observation, we formulate incremental learning as recursive estimation of class-conditional Gaussian densities in foundation embeddings. Each class is represented by its sufficient statistics, which can be updated incrementally without backpropagation or exemplar storage. While Gaussian density estimation is classical, we show that foundation feature spaces fundamentally alter its effectiveness, making second-order statistical modeling sufficient for competitive incremental learning. Despite its simplicity, the proposed approach achieves performance on CIFAR-100 and ImageNet-R within a narrow margin of the best recent prompt-based methods, while requiring significantly lower computational cost and providing a principled probabilistic interpretation. These results suggest that, in the foundation model era, incremental learning may be more naturally addressed through density estimation in stable representation spaces rather than continual parameter adaptation.

Keywords: Class-Incremental Learning · Foundation Models · Gaussian Density Estimation

1 Introduction

Class-Incremental Learning (CIL) aims to extend a recognition system with new classes while maintaining performance on previously learned categories. The dominant view treats this problem as one of parameter interference: learning new classes requires updating network weights, which may overwrite previously acquired knowledge. Consequently, most existing methods attempt to control parameter updates through regularization strategies, replay buffers, or architectural adaptations. Although effective, these approaches share a common premise: incremental learning must be achieved through careful optimization of model parameters.

Interestingly, this premise contrasts with the organization of the human visual system. In biological vision, early cortical areas develop feature extraction mechanisms during the early post-natal stages and subsequently provide relatively stable perceptual representations. Yet humans retain the ability to learn new object categories throughout life. Higher-level cognitive processes associate these stable visual representations with semantic categories.

A similar structural separation has emerged in the era of large-scale visual foundation models. Encoders trained on massive datasets produce highly discriminative representations that remain stable when frozen. Under such stationary feature spaces, representation drift—the main source of catastrophic forgetting in neural networks—is eliminated by construction. From this perspective, class-incremental learning can be viewed as the problem of modeling class distributions in a fixed embedding space rather than adapting feature extractors.

In this work, we revisit a classical idea in this modern context: modeling class-conditional densities using Gaussian statistics. Although Gaussian classifiers have long been studied in pattern recognition, their effectiveness was historically limited by unstable or weak representations. Foundation models fundamentally change this landscape. In their embedding spaces, simple second-order statistical models provide a strong inductive bias that enables effective incremental recognition.

Building on this observation, we formulate the CIL as a recursive estimation of class-conditional Gaussian densities in foundation embeddings. Each class is represented by its sufficient statistics, which can be updated incrementally without backpropagation, replay buffers, or exemplar storage. This formulation yields a simple probabilistic classifier with negligible computational overhead.

Despite its conceptual simplicity, the proposed approach achieves performance on CIFAR-100 and ImageNet-R within a narrow margin of the best recent methods, which rely on complex prompt-based adaptations of transformer architectures. These results suggest that, in the foundation model era, the central challenge of incremental learning may lie less in preventing weight forgetting and more in exploiting the statistical structure of stable representation spaces. While density estimation is classical, its application as a total replacement for architectural adaptation in CIL is a novel paradigm enabled specifically by the emergent properties of foundation embeddings

In summary, the main contributions of this work are:

- **Reformulation.** We recast CIL as recursive density estimation in frozen foundation feature spaces, shifting the focus from parameter preservation to statistical modeling.
- **Probabilistic modeling.** We demonstrate that maintaining class-conditional first- and second-order sufficient statistics is sufficient for competitive incremental performance in modern embeddings.
- **Efficiency and privacy.** We provide a computationally-efficient and privacy-preserving implementation requiring no exemplar storage, no backpropagation, and no prompt parameters.

– **Competitive performance.** We achieve performance on CIFAR-100 and ImageNet-R within a narrow margin of the strongest recent methods, while introducing only minimal computational overhead beyond the backbone.

2 Related Work

Regularization-based methods. To maintain model stability across sequential tasks, these methods limit updates to parameters identified as important for prior learning. These approaches typically introduce regularization terms in the training objective that penalize changes to such parameters. Representative strategies include: (1) slowing updates to parameters estimated to be important for prior tasks [2, 17]; (2) restricting learning in neurons that are critical for earlier tasks [16]; and (3) introducing additional loss terms based on knowledge distillation to preserve previous task outputs while learning new tasks [19].

Rehearsal-based methods. To preserve the model's past performance, these methods reuse samples from previously learned tasks while training on the current task. Some approaches store a small set of exemplars from past tasks and replay them during training [23]. Others maintain a limited memory buffer containing samples and class-related statistics from previous tasks [3]. Another strategy uses stored samples to constrain gradient updates during training on new tasks, reducing interference with previously learned knowledge [5, 20]. A complementary approach is generative replay, where a generative model is trained to produce samples or feature representations from previous tasks. In this case, the generator itself must also mitigate catastrophic forgetting, often using replay or distillation-based techniques [22, 25, 30].

Architecture-based methods. These methods mitigate catastrophic forgetting by modifying the network architecture to increase or reorganize its learning capacity across tasks. Some approaches [13, 14] identify and preserve parameters that are important for previous tasks—often through pruning or gradient masking—so that only a subset of weights is updated for the current task. When the available capacity becomes insufficient, the architecture can be expanded to accommodate new knowledge. In [31], rehearsal is combined with an expandable super-feature extractor that grows as new tasks arrive, followed by a pruning step to maintain efficiency. Similarly, [32] proposes expanding the network when necessary while selectively retraining only the subset of parameters from the previous network that are relevant to the current task.

Prompt-based methods. The emergence of foundation models, particularly Vision Transformers (ViTs), has enabled parameter-efficient fine-tuning (PEFT) strategies based on learned prompts injected into frozen backbones. L2P [28] introduces a prompt pool, where a subset of prompts is selected based on the similarity between the image embeddings and the learned keys. DualPrompt [27] extends this idea by combining task-specific prompts with shared general prompts. PGP [15] further constrains gradient updates to reduce interference with previous tasks, leveraging prompts to avoid explicit task identifiers. CODA-Prompt [1] proposes a fully differentiable prompt construction mechanism, generating prompts

as attention-weighted combinations of learned components. CPrompt [9] addresses training–testing inconsistencies by exposing task prompts to all previously learned classifiers and introducing prompt sampling during training. ConvPrompt [24] generates task-specific prompts via convolution over a shared embedding matrix that encodes prior knowledge. PromptFusion [7] separates stability and plasticity using two prompt modules: a Stabilizer that preserves prior knowledge and a Booster that adapts to new tasks. APT [6] simplifies prompt usage by directly injecting learnable key-value tokens into each transformer layer, avoiding explicit prompt selection and reducing inference overhead. Finally, RainbowPrompt [12] enhances representational diversity through a prompt-evolving mechanism that progressively integrates knowledge from past and current tasks. Notably, the current CIL state-of-the-art landscape is dominated by prompt-based methods.

Discussion and Motivation. As just mentioned, state-of-the-art CIL approaches predominantly rely on parameterized prompt mechanisms that expand or adapt transformer architectures across incremental phases. Although empirically effective, such strategies introduce additional parameters and optimization procedures whose memory and computational complexity grow with the number of tasks or classes. Moreover, their interaction with frozen foundation representations remains only partially understood from a statistical perspective, since adaptation occurs at the architectural level rather than at the level of distributional modeling. In contrast, we investigate whether incremental learning in the foundation model regime can be formulated without architectural modification. We propose to treat CIL as recursive estimation of class-conditional distributions in a fixed embedding space, thereby replacing continual parameter adaptation with principled statistical modeling. This shift allows us to examine the extent to which the structure of modern foundation embeddings renders second-order density estimation sufficient for competitive incremental performance.

Recent work has explored classifier adaptation on top of frozen feature extractors. SLCA [33] and related approaches investigate how to incrementally adjust the classification layer of pre-trained models while maintaining stable representations. Analytic Class-Incremental Learning (ACIL) [34] further proposes closed-form updates for linear classifiers, enabling efficient incremental learning without backpropagation. Another line of work studies prototype-based classifiers in frozen feature spaces. FeCAM [10] models the class covariance structure and performs classification using a Mahalanobis distance metric, highlighting the importance of heterogeneous feature distributions. In contrast to these approaches, which remain fundamentally discriminative or metric-based, our method adopts a probabilistic perspective and formulates class-incremental learning as recursive estimation of class-conditional densities in a stationary embedding space.

3 Proposed Method

Our CIL framework is built upon a simple yet structurally motivated decomposition: (i) a frozen feature extractor based on a visual foundation model, and (ii) a probabilistic classification layer that models each object category through

an independent Gaussian distribution in the embedding space. Given an input image, the feature extractor maps it to a fixed high-dimensional representation. For each class, we maintain a class-conditional multivariate Gaussian characterized by its mean vector and covariance matrix, estimated incrementally from observed samples. Classification is performed according to the maximum likelihood principle, assigning the input to the class whose Gaussian model yields the highest likelihood under the stationary embedding representation. This modular separation between representation learning and statistical modeling enables incremental updates without backpropagation or parameter adaptation: adding a new class requires only estimating its distribution in the fixed feature space, while previously learned classes remain unaffected at the representation level.

3.1 Foundation Feature Space

A fundamental assumption of our framework is the availability of a pre-trained and frozen feature extractor. The representation model must be trained prior to the initialization of the incremental recognition system and remain fixed during all subsequent class-incremental updates. Moreover, the resulting embedding space must be sufficiently general and discriminative to characterize not only the classes observed during its pre-training phase, but also novel categories introduced over time.

This design is inspired by the organization of the human visual system. Early cortical areas, such as V1 and V2, develop stable feature detectors through exposure to diverse visual stimuli during early development [26]. Once established, these perceptual representations remain largely stable, while higher-level regions enable the acquisition of new semantic categories throughout life. Analogously, our framework assumes a stationary perceptual embedding space upon which incremental category learning is performed.

In practice, such representations require training on large-scale and diverse image corpora. Modern visual foundation models satisfy this requirement by leveraging supervised or self-supervised learning on massive datasets, yielding highly transferable and structured feature spaces. In our experiments, we employ Vision Transformer (ViT) [8] and DINOv2 [21] models as representative foundation encoders. These models are used exclusively as frozen feature extractors, without any fine-tuning during incremental learning.

3.2 Feature Representation

Given an input image $x \in \mathcal{X}$, we extract a fixed-dimensional embedding using a pretrained visual foundation model:

$$\mathbf{z} = \mathcal{F}(x) \in \mathbb{R}^d. \tag{1}$$

The encoder $\mathcal{F}$ remains frozen during training and inference. All learning is performed directly in the feature space $\mathbb{R}^d$.

Let $\mathcal{Y} = \{1, \ldots, C\}$ denote the set of classes.

3.3 Likelihood-Based Classification

We adopt a generative modeling framework and assume class-conditional Gaussian feature distributions:

$$p(\mathbf{z} \mid y = c) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{2}$$

where $\boldsymbol{\mu}_c \in \mathbb{R}^d$ and $\boldsymbol{\Sigma}_c \succ 0$.

Classification is performed via maximum a posteriori (MAP) inference:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\mathbf{z} \mid y = c) + \log \pi_c \right], \tag{3}$$

where $\pi_c = p(y = c)$ denotes the class prior.

Using the Gaussian density, the decision rule is equivalent to

$$\hat{y} = \arg\min_{c} \left[(\mathbf{z} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| - 2 \log \pi_c \right]. \tag{4}$$

This yields a quadratic decision function in the feature space.

3.4 Covariance Modeling

We consider three covariance parameterizations.

Full covariance. Each class is assigned an independent covariance matrix:

$$\boldsymbol{\Sigma}_c \in \mathbb{R}^{d \times d}. \tag{5}$$

Diagonal covariance. We restrict covariance matrices to be diagonal:

$$\boldsymbol{\Sigma}_c = \mathrm{diag}(\boldsymbol{\sigma}_c^2). \tag{6}$$

The quadratic term simplifies to

$$(\mathbf{z} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c) = \sum_{i=1}^{d} \frac{(z_i - \mu_{c,i})^2}{\sigma_{c,i}^2}. \tag{7}$$

Shared covariance. We impose a common covariance across classes:

$$\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}, \quad \forall c. \tag{8}$$

The discriminant function becomes

$$g_c(\mathbf{z}) = \mathbf{z}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c - \frac{1}{2} \boldsymbol{\mu}_c^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c + \log \pi_c. \tag{9}$$

3.5 Mixture of Gaussians Extension

To capture multimodal feature distributions, we extend the class-conditional model to a mixture:

$$p(\mathbf{z} \mid y = c) = \sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}), \tag{10}$$

where $\alpha_{c,k} \geq 0$ and $\sum_k \alpha_{c,k} = 1$.

The MAP decision rule becomes

$$\hat{y} = \arg\max_c \left[\log \left(\sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}) \right) + \log \pi_c \right]. \tag{11}$$

3.6 Parameter Estimation and Incremental Updates

Given training data $\{(\mathbf{z}_i, y_i)\}_{i=1}^N$, the parameters are estimated using maximum likelihood. Mixture parameters are estimated using the EM algorithm. All sufficient statistics admit closed-form incremental updates (except for Mixtures). Thus, class expansion requires only summary statistics and no storage of past samples.

4 Experiments

4.1 Experimental Setup

Datasets and Protocol. We follow the standard class-incremental evaluation protocol. Experiments are conducted on CIFAR-100 [18] and ImageNet-R [11]. CIFAR-100 consists of 100 object categories, while ImageNet-R contains 200 ImageNet classes rendered in diverse artistic and non-photorealistic styles, inducing substantial distributional variability. For CIFAR-100, we adopt the commonly used 10-task and 20-task class-incremental splits, where disjoint subsets of classes are introduced sequentially and task identity is not provided at test time (single-head evaluation). For ImageNet-R, we follow a similar protocol by partitioning the 200 classes into 10 and 20 incremental tasks. In all settings, models are evaluated after each incremental phase on the union of all classes observed so far.

Joint Cross-Dataset Benchmark. In addition, we introduce a cross-dataset incremental setting combining CIFAR-100 and ImageNet-R within a unified class-incremental stream. The 100 CIFAR-100 classes are divided into 10 tasks, followed by 10 tasks constructed from ImageNet-R, resulting in 20 sequential phases. This setting evaluates the ability to incrementally incorporate semantically and distributionally distinct categories within a single stationary embedding space.

Evaluation Metrics. We report the standard metrics for class-incremental learning: Average Accuracy and Forgetting (see definition in [4]).

Methods. We analyze four variants of our Gaussian-based methods: Full-Cov, Diag-Cov and Shared-Cov, which use full, diagonal and shared covariance matrices, respectively, and MoG-4, which uses a Mixture of Gaussian model, with up to 4 Gaussians. We compared these methods with the following prompt-based methods: L2P [28], DualPrompt [27], DualPrompt-PGP [15], CODA-Prompt [1], CPrompt [9], ConvPrompt [24], RainbowPrompt [12] and APT [6]. Due to rank limitations in the Full-Cov model, covariances are regularized using diagonal loading (adding 10^{-6} to the diagonal) and by enforcing a minimum eigenvalue of 10^{-6}. Also, ImageNet-R class samples are reduced to 40 per class (balanced) when training Full-Cov.

Visual Foundation Models. We evaluated our method using different foundation models as features extractors: (1) DINOv2-B/14 [21], (2) ViT-B/16 pre-trained on ImageNet-21k [8], and (3) ViT-Medium (ViT-M/16) [29].

4.2 Class-incremental classification

Table 1 reports the performance of the four variants of the proposed Gaussian-based classifier together with the prompt-based continual learning methods, using the performances reported in the corresponding papers. In these experiments, DINOv2 is used as the backbone feature extractor for the Gaussian-based classifiers. Table 2 further analyzes the impact of the backbone architecture by reporting the performance of the Diag-Cov and Full-Cov variants using different foundation models.

A first observation from Table 1 is that the proposed Gaussian-based methods achieve performance within a narrow margin of the strongest prompt-based approaches, despite relying on a significantly simpler classification mechanism. Unlike prompt-based methods, which require additional trainable parameters and optimization during incremental learning, the proposed approach performs classification through direct probabilistic modeling in the feature space.

Another notable property is that, unlike the prompt methods, the accuracies obtained are deterministic with respect to the class presentation order. Since the method maintains only sufficient class-conditional statistics and does not involve parameter optimization during incremental updates, the resulting classifier is independent of the specific ordering of classes. Consequently, the reported A10 and A20 accuracies for each method remain identical across runs.

Regarding forgetting, the proposed framework does not exhibit forgetting in the conventional sense, as previously learned class statistics are never modified when new classes are introduced. The measured forgetting values therefore primarily reflect the increased classification difficulty as the number of competing classes grows, rather than degradation of previously learned representations.

Table 2 highlights the dependence of classification performance on the quality of the underlying feature space. As expected, stronger foundation models such as DINOv2 produce more discriminative embeddings, leading to improved accuracy. This observation suggests that the effectiveness of the proposed probabilistic modeling framework is closely tied to the representational quality of the backbone,

Table 1: Results on CIFAR-100 and ImageNet-R. A_N and F_N represent the average accuracy and forgetting for the N-task setting, respectively. Note: Forgetting in our method reflects statistical interference rather than parameter overwriting.

Method	CIFAR-100				ImageNet-R			
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{20}(\uparrow)$	$F_{20}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{20}(\uparrow)$	$F_{20}(\downarrow)$
L2P [28]	82.76 ±1.17	7.86 ±0.39	77.95 ±0.72	9.88 ±0.30	63.49 ±0.40	6.85 ±0.42	59.38 ±0.50	5.89 ±0.36
DualPrompt [27]	85.07 ±0.49	5.57 ±0.20	80.49 ±0.31	8.84 ±0.68	68.50 ±0.52	5.14 ±0.18	63.21 ±0.49	5.28 ±0.45
DualPrompt-PGP [15]	86.92 ±0.05	5.35 ±0.19	83.74 ±0.01	7.91 ±0.15	69.34 ±0.05	4.53 ±0.04	64.75 ±0.38	6.04 ±0.15
CODA-Prompt [1]	87.00 ±0.38	4.78 ±0.24	82.15 ±0.17	6.33 ±0.23	74.24 ±0.56	4.92 ±0.21	70.86 ±0.42	6.87 ±0.25
CPrompt [9]	87.83 ±0.37	4.88 ±0.79	84.66 ±0.13	5.69 ±0.06	76.71 ±0.61	4.66 ±0.93	74.45 ±0.25	4.98 ±0.21
ConvPrompt [24]	88.87 ±0.33	4.75 ±0.15	87.37 ±0.13	5.16 ±0.01	77.86 ±0.25	4.33 ±0.24	75.10 ±0.39	4.10 ±0.29
RainbowPrompt [12]	89.86 ±0.11	3.44 ±0.26	90.15 ±0.05	3.75 ±0.23	79.09 ±0.13	3.90 ±0.23	78.36 ±0.47	3.44 ±0.29
APT [6]	88.88 ±0.65	3.47 ±0.55	86.37 ±0.46	4.57 ±0.88	79.40 ±0.47	4.38 ±0.46	76.22 ±0.04	6.91 ±0.33
Diag-Cov (ours)	86.07 ±0.00	4.30 ±0.12	86.07 ±0.00	7.14 ±0.48	77.12 ±0.00	7.44 ±0.17	77.12 ±0.00	6.92 ±2.33
Full-Cov (ours)	88.48 ±0.00	4.18 ±0.00	88.48 ±0.00	5.64 ±0.36	73.55 ±0.00	4.69 ±0.20	73.55 ±0.00	7.98 ±2.02
MoG4 (ours)	88.87 ±0.00	4.03 ±0.00	88.87 ±0.00	5.28 ±0.37	75.07 ±0.00	5.07 ±0.05	75.07 ±0.00	7.68 ±1.5
Shared-Cov (ours)	88.56 ±0.00	3.76 ±0.11	88.56 ±0.00	5.73 ±0.32	77.06 ±0.01	3.97 ±0.30	77.06 ±0.00	7.74 ±1.29

Table 2: Comparison of different backbones (DINOv2-B/14, ViT-M/16, ViT-B/16) using Diagonal and Full Covariance modeling on CIFAR-100 and ImageNet-R.

Method	Backbone	CIFAR-100		ImageNet-R	
		$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
Diag-Cov	DINOv2-B/14	86.07 ±0.00	4.30 ±0.12	77.12 ±0.00	7.44 ±0.17
	ViT-M/16	81.41 ±0.00	5.28 ±0.13	66.11 ±0.00	4.79 ±0.14
	ViT-B/16	81.03 ±0.00	4.57 ±0.41	43.08 ±0	6.41 ±0.19
Full-Cov	DINOv2-B/14	88.48 ±0.00	4.18 ±0.00	73.55 ±0.00	4.69 ±0.20
	ViT-M/16	82.42 ±0.00	5.25 ±0.01	66.44 ±0.01	5.21 ±0.14
	ViT-B/16	80.40 ±0.00	5.38 ±0.36	45.72 ±0.00	7.05 ±0.12

and may further benefit from future advances in large-scale visual representation learning.

Finally, Table 3 reports the performance of the four variants of the proposed Gaussian-based classifier on the combined benchmark constructed from CIFAR-100 and ImageNet-R. This setting is particularly challenging, as it involves 300 classes originating from two datasets with substantially different visual characteristics. Despite this increased diversity and scale, the strongest variants of the proposed classifier achieve accuracies above 80%, demonstrating that Gaussian modeling in foundation feature spaces remains effective even when the number and heterogeneity of classes increase substantially.

Table 3: Results on the Joint Dataset Benchmark.

Method	Joint Dataset			
	$A_{10}(\uparrow)$		$F_{10}(\downarrow)$	
Diag-Cov (ours)	78.46	±0.00	5.33	±0.24
Full-Cov (ours)	82.93	±0.00	4.51	±0.00
MoG4 (ours)	82.36	±0.00	4.68	±0.00
Shared-Cov (ours)	81.20	±0.00	5.03	±0.28

4.3 Analysis of Gaussianity assumptions

In this section, the CIFAR-100 training data is used to evaluate whether the Full-Cov and Diag-Cov models provide appropriate representations of the data. Additionally, the structure of the DINOv2 feature space is analyzed to better understand the validity of the Gaussian modeling assumptions.

Analysis of the DINOv2 feature space. The DINOv2 feature space has dimensionality 768. To assess the amount of information lost under dimensionality reduction, we analyze the spectral decay of the feature covariance matrix. Specifically, Principal Component Analysis (PCA) was performed on the DINOv2 features extracted from both the CIFAR-100 training set and the ImageNet-21k dataset. As shown in Figure 1, the eigenvalues exhibit a steep decay in approximately the first 50 principal components, followed by a much more gradual decline. This behavior indicates that a substantial portion of the variance is captured by the leading components, while the remaining components encode progressively finer-grained information. Consequently, restricting the representation to a small number of principal components inevitably results in non-negligible information loss.

Gaussianity of the individual components. To evaluate the assumption that each feature component is approximately Gaussian within each class, we computed the kurtosis of each component for every class in the CIFAR-100 training set. Since the DINOv2 feature representation has 768 dimensions and CIFAR-100 contains 100 classes, this procedure resulted in a total of $76,800$ kurtosis values. The distribution of these kurtosis values is shown in Figure 2a. It can be seen that the vast majority of the values lie within the interval $[-1, 1]$. This range is commonly regarded as indicative of a moderate deviation from normality, suggesting that, for most classes and components, the Gaussianity assumption is reasonable. Consequently, modeling the class-conditional feature distributions as Gaussian appears to be a justified approximation.

Assessment of the Covariance Models. To assess whether the Full-Cov and Diag-Cov Gaussian models are appropriate, we analyze the distribution of the squared Mahalanobis distances on the CIFAR-100 dataset. If the data follows an n-dimensional Gaussian distribution with full-rank covariance matrix, the squared Mahalanobis distance (computed with respect to the true class mean and covariance) follows a chi-square distribution with n degrees of freedom. As shown in Figure 2b, the empirical distribution of the squared Mahalanobis distances under the Full-Cov model deviates significantly from the theoretical χ^2 distribution. This discrepancy arises because the estimated covariance matrices are rank-deficient. Specifically, when a class contains N training samples, the empirical covariance matrix has a rank of at most $N - 1$. Consequently, only $N - 1$ eigenvalues are strictly positive. The remaining eigenvalues are numerically regularized by setting them to a small constant value 10^{-6} to ensure invertibility. Although the Full-Cov model performs well in practice, its rank deficiency introduces instability. In particular, when the number of training samples varies across classes, the effective dimensionality of the covariance estimates differs, which can degrade classification performance in imbalanced settings. Developing

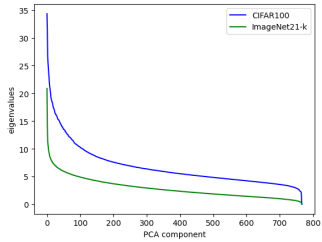


Fig. 1: PCA Eigenvalues decomposition of CIFAR-100 and ImageNet21-k features.

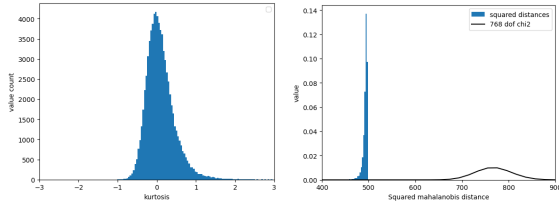


Fig. 2: (a) Kurtosis histogram. (b) Squared Mahalanobis distances histogram for the full-covariance model and the corresponding theoretical chi-square distribution.

a principled strategy for handling low-rank covariance estimation remains an important direction for future work.

As shown in Figure 3a, the empirical distribution of the squared Mahalanobis distances under the Diag-Cov model differs noticeably from the corresponding theoretical chi-square distribution. In particular, the empirical distribution appears to be more dispersed than expected. However, when the same analysis is restricted to a single feature component (e.g., the first component), the resulting squared Mahalanobis distances closely follow a chi-square distribution with one degree of freedom, as shown in Figure 3b. These results suggest that the discrepancy observed in the multidimensional case arises from the diagonal covariance assumption. Although each individual feature is approximately Gaussian within each class —as supported by the kurtosis analysis— the joint distribution exhibits non-negligible correlations across components. By construction, the Diag-Cov model ignores these off-diagonal covariance terms, which leads to a mismatch between the empirical and theoretical chi-square distributions in the multivariate setting. Therefore, although the Diag-Cov model avoids the rank-deficiency issues associated with the full-covariance model, it sacrifices representational capacity by failing to model inter-feature dependencies.

Inter-Class Similarity of Full-Covariance Gaussian Models. In addition to assessing the validity of the Gaussian assumption, we analyze the similarity between the class-conditional Gaussian distributions obtained with the Full-Cov model. To this end, we consider 100 single-class classification tasks, one per class. For each task, a full-covariance Gaussian model is estimated using the corresponding training data, and its classification accuracy is evaluated on the test set. To quantify inter-class similarity, we compute pairwise Kullback–Leibler (KL)

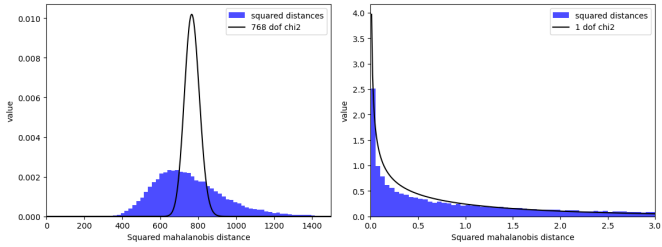


Fig. 3: (a): Squared Mahalanobis distances for the Diag-Cov model and the theoretical chi-square distribution. (b): Squared Mahalanobis distances for the one-dimensional model and the corresponding chi-square distribution.

divergences between the estimated Gaussian distributions, and summarize them using aggregate statistics (e.g., average and minimum pairwise KL divergence). These metrics are analyzed as a function of the number of classes considered. As shown in Figure 4, both the average pairwise KL divergence and the average classification accuracy decrease gradually as the number of classes increases. This indicates that, on average, the class-conditional Gaussian distributions remain distinguishable even as the classification problem becomes more complex. However, the minimum pairwise KL divergence decreases sharply. This suggests the existence of specific pairs of classes whose Gaussian representations are highly similar, despite the overall separation between classes. A comparable trend is observed in the test accuracy: while the average performance degrades smoothly, certain class combinations lead to more pronounced drops.

Finally, it is important to note that the performance of the Full-Cov model cannot be trivially improved through linear transformations of the feature space. The Mahalanobis distance is invariant under non-singular linear transformations of the data [35], implying that operations such as whitening or other invertible linear mappings do not alter the induced decision rule. Therefore, the behavior observed in Figure 4 cannot be mitigated simply by applying linear preprocessing to the features.

Conclusions on Gaussian representations. Feature representations appear approximately Gaussian for most classes; however, the Full-Cov model's reliance on the full second-order structure leads to low-rank covariance estimates in cases where the sample size is smaller than the feature dimensionality. Although regularization alleviates this issue and yields good empirical performance, the effective covariance rank depends on the available samples, making the model sensitive to class imbalance. In contrast, the Diag-Cov model avoids rank deficiency and is more robust in unbalanced settings. However, it ignores inter-feature correlations and therefore cannot capture the full joint feature distribution. As a result, while the marginal distributions are approximately Gaussian, important dependencies across components are not modeled. Overall, Gaussian modeling appears justified, but better strategies for covariance estimation under limited and imbalanced data are still needed, particularly methods that mitigate rank deficiency while preserving inter-feature correlations.

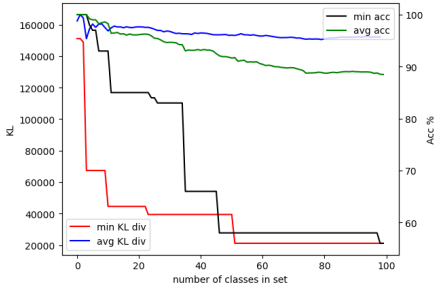


Fig. 4: Minimum and average pairwise KL divergence between class Gaussians, together with minimum and average task accuracy, as a function of the number of classes.

4.4 Computational requirements

The evaluation of Gaussian models is computationally efficient, as it mainly involves matrix–vector multiplications. For the Diag-Cov model, the total computational cost amounts to approximately 0.4608 MFLOPs on CIFAR-100 and 0.9216 MFLOPs on ImageNet-R when evaluated across all tasks. In the case of the Full-Cov model, the corresponding cost is approximately 0.1183 GFLOPs on CIFAR-100 and 0.2367 GFLOPs on ImageNet-R. These computational requirements are negligible compared to the approximately 17.56 GFLOPs required for a forward pass of a DINOv2 ViT-B/14 backbone. Therefore, the Gaussian classification stage introduces only a marginal computational overhead relative to feature extraction.

4.5 Incremental Gaussian Model Update

To illustrate the ability of the proposed classifier to estimate Gaussian models incrementally, we analyze a scenario where samples from a new class are introduced progressively. After learning an initial set of nine classes, an additional class is introduced. The training samples of this new class are incorporated incrementally in steps of 10% of the available data. Figure 5(a) shows the results for CIFAR-100, while Figure 5(b) presents the corresponding results for ImageNet-R. For CIFAR-100, the Diag-Cov model achieves relatively strong performance with only 10% of the samples, reflecting the small number of parameters required to estimate diagonal covariance matrices. However, its performance quickly saturates as additional samples are incorporated. In contrast, the Full-Cov model improves more gradually, requiring approximately 50–70% of the samples before surpassing the Diag-Cov model. For ImageNet-R, the analysis is constrained by the limited number of training samples per class. In particular, only 40 samples per class—the minimum across classes—can be used to estimate the Full-Cov model due to rank limitations of the covariance matrix. Under this setting, both the Diag-Cov and Full-Cov models exhibit slower performance improvements as additional samples are incorporated.

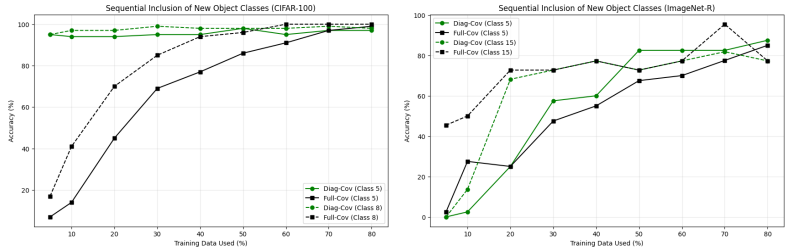


Fig. 5: Accuracy of a new class as samples arrive incrementally: (a) CIFAR-100, (b) ImageNet-R.

5 Conclusion

We revisited class-incremental learning with modern foundation models, arguing that frozen, discriminative representations shift the challenge from parameter adaptation to statistical modeling. We formulated CIL as recursive estimation of class-conditional Gaussian densities, maintaining only first- and second-order sufficient statistics. This enables incremental learning without backpropagation, replay buffers, or prompt-based architectural adaptation.

Empirically, this simple formulation achieves competitive performance on CIFAR-100 and ImageNet-R—within a narrow margin of the strongest recent methods—while substantially reducing computational overhead. These results suggest foundation model embeddings fundamentally alter the effectiveness of classical density estimation, where second-order statistics provide a strong, stable inductive bias. The analysis in 4.3 also shows that the Gaussianity assumption is reasonable, despite the difficulty of estimating covariance matrices with limited samples per class.

Beyond the specific Gaussian model, our findings point toward a broader research direction: treating continual learning as principled distribution modeling in stable representation spaces. Future work may explore richer generative models, structured covariance estimation, uncertainty calibration, and extensions to open-world or domain-incremental settings. More generally, as foundation models improve, the role of continual learning may shift from protecting parameters to exploiting the statistical geometry of learned feature spaces.

We believe this perspective opens new directions for developing efficient continual learning systems that leverage the statistical structure of large-scale foundation model embeddings.

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